

Q1 - 25 July - Shift 2

If the mean deviation about median for the number 3, 5, 7, $2k$, 12, 16, 21, 24 arranged in the ascending order, is 6 then the median is

- (A) 11.5 (B) 10.5
(C) 12 (D) 11

Space for your notes:

Q2 - 26 July - Shift 2

The mean and standard deviation of 40 observations are 30 and 5 respectively. It was noticed that two of these observations 12 and 10 were wrongly recorded. If σ is the standard deviation of the data after omitting the two wrong observations from the data, then $38\sigma^2$ is equal to _____.

Space for your notes:

Q3 - 27 July - Shift 1

The mean and variance of 10 observations were calculated as 15 and 15 respectively by a student who took by mistake 25 instead of 15 for one observation. Then, the correct standard deviation is _____.

Space for your notes:

Q4 - 29 July - Shift 1

Questions

MathonGo

Let the mean and the variance of 20 observations $x_1, x_2, \dots, x_{20}$ be 15 and 9, respectively. For $\alpha \in \mathbb{R}$, if the mean of $(x_1 + \alpha)^2, (x_2 + \alpha)^2, \dots, (x_{20} + \alpha)^2$ is 178, then the square of the maximum value of α is equal to _____.

Space for your notes:

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Q1 (D)

Q2 (238)

Q3 (2)

Q4 (4)

Q1 (D)

(D)

Q2 (238)

$$\text{Wrong mean} = \mu_1 = 30$$

$$\text{Wrong S.D} = \sigma_1 = 5$$

$$\frac{\sum x_i}{40} = 30$$

$$\Rightarrow \sum x_i = 1200$$

$$\sigma_1^2 = 25$$

$$\Rightarrow \frac{\sum x_i^2}{40} - 30^2 = 25$$

$$\Rightarrow \sum x_i^2 = 925 \times 40 = 37000$$

$$\text{New sum} = \sum x'_i = 1200 - 10 - 12 = 1178$$

$$\text{New mean} = \mu'_1 = \frac{1178}{38} = 31$$

$$\text{New } \sum x_i^2 = 37000 - (10)^2 - (12)^2 = 36756$$

$$\text{New S.D, } \sigma'_1 = \sqrt{\frac{36756}{38} - (31)^2} = \sigma$$

$$36756 - (31)^2 \times 38 = 38\sigma^2$$

$$\Rightarrow 38\sigma^2 = 238$$

Q3 (2)

$$6^2 = \frac{\sum x_i^2}{10} - (\bar{x})^2 = 15$$

$$\Rightarrow \sum_{i=1}^{10} x_i = 150$$

$$\Rightarrow \sum_{i=1}^9 x_i + 25 = 150$$

$$\Rightarrow \sum_{i=1}^9 x_i = 125$$

$$\Rightarrow \sum_{i=1}^9 x_i + 15 = 140$$

$$\text{Actual mean} = \frac{140}{10} = 14 = \bar{x}_{\text{new}}$$

$$\sum_{i=1}^9 \frac{x_i^2 + 25^2 - 15^2}{10} = 15$$

$$\Rightarrow \sum_{i=1}^9 x_i^2 + 625 = 2400$$

$$\sum_{i=1}^9 x_i^2 = 1775$$

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Statistics
Hints and Solutions

$$\sum x_i + 15 = 2000 = \left(\sum x_i \right)_{actual}$$

$$6_{actual}^2 = \frac{\left(\sum x_i^2 \right)_{actual} - (\bar{x}_{new})^2}{10}$$

$$= \frac{2000}{10} - 14^2$$

$$= 200 - 196 = 4$$

$$(S.D)_{actual} = 6 = 2$$

Q4 (4)

$$\sum x_1 = 15 \times 20 = 300 \quad \dots(i)$$

$$\frac{\sum x_1^2}{20} - (15)^2 = 9 \quad \dots(ii)$$

$$\sum x_1^2 = 234 \times 20 = 4680$$

$$\frac{\sum (x_1 + \alpha)^2}{20} = 178 \Rightarrow \sum (x_1 + \alpha)^2 = 3560$$

$$\Rightarrow \sum x_1^2 + 2\alpha \sum x_1 + \sum \alpha^2 = 3560$$

$$4680 + 600\alpha + 20\alpha^2 = 3560$$

$$\Rightarrow \alpha^2 + 30\alpha + 56 = 0$$

$$\Rightarrow (\alpha + 28)(\alpha + 2) = 0$$

$$\alpha = -2, -28$$

Square of maximum value of α is 4